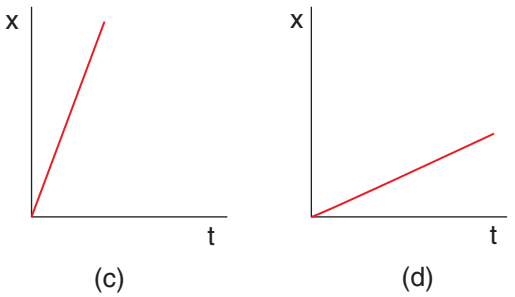
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The slope in the position – time graph will give the speed of the particle.

In the graph (a) slope is zero. Graph (c) has higher slope than graphs (b) and (d). So we can arrange the speeds in ascending order as

$$v_a < v_d < v_b < v_c$$

SOLVED EXAMPLE UNIT-3

1. A body of mass 100 kg is moving with an acceleration of 50 cm s^{-2}. Calculate the force experienced by it.

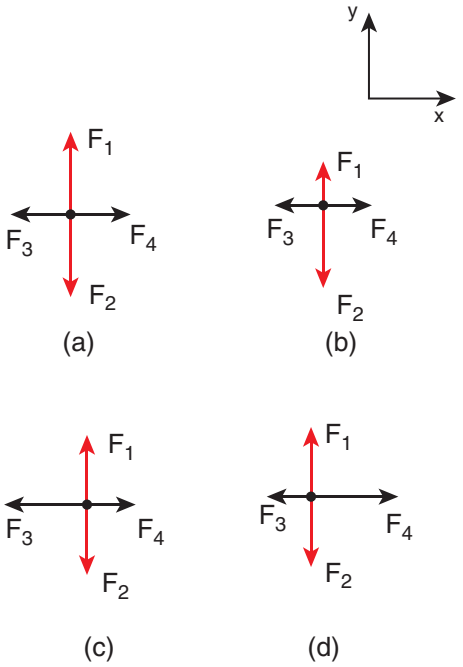
Solution:

Mass m = 100 kg
Acceleration a = 50 cms^{-2} = 0.5 ms^{-2}
Using Newton's second law,

$$F = ma$$
$$F = 100 \text{ kg} \times 0.5 \text{ m s}^{-2} = 50 \text{ N}$$

2. Identify the free body diagram that represents the particle accelerating in positive *x* direction in the following.

The relative magnitude of forces should be indicated when the free body diagram for mass m is drawn.



Case (a):
The forces F_1 and F_2 have equal length but opposite direction. So net force along y-direction is zero. Since the force is zero, acceleration is also zero along Y-direction (Newton's second law). Similarly in the x direction, F_3 and F_4 have equal length and opposite in direction. So net force is zero in the x direction. So there is no acceleration in x direction.

Case (b):
The forces F_1 and F_2 are not equal in length and act opposite to each other. The figure (b) shows that there are unbalanced forces along the y-direction. So the particle has acceleration in the -y direction. The forces F_3 and F_4 are having equal length and act in opposite directions. So there is no net force along the x direction. So the particle has no acceleration in the x direction.

Case (c):
The forces F_1 and F_2 are equal in magnitude and act opposite to each